Gait Detection For Human Posture Analysis

A project report submitted in partial fulfillment

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Bachelor of Technology

in

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Declaration

I hereby declare that the report titled ***Gait Analysis For Human Posture Analysis*** submitted by me to the School of Electronics Engineering, Vellore Institute of Technology, Chennai in partial fulfillment of the requirements for the award of **Bachelor of Technology** in **Electronics and Communication Engineering** is a bona-fide record of the work carried out by me under the supervision of ***Dr. S. Sivakumar*** **Associate Professor Senior**. I further declare that the work reported in this report, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or University.

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Abstract

Sedentary lifestyles have caused posture degradation to become a public health concern that adversely contributes to chronic musculoskeletal conditions, such as back pain, joint issues, and loss of mobility. This paper consists of a real-time posture and gait analysis system based on OpenCV and Mediapipe for conducting real-time image processing and landmark detection. A custom dataset of four classes for posture-good sitting, bad sitting, good standing, and bad standing-is used to train a 1-D CNN. The architecture of the CNN will capture the spatial relationship between key body landmarks, leading to accurate classification.

The system includes IoT capabilities and the use of an ESP32 microcontroller that is connected to a vibrating motor for providing instant haptic feedback whenever poor posture is detected. It will enable users to correct posture issues in real time, thus effectively addressing posture-related challenges. The model ensures performance with robust accuracy and efficiency in different applications with real-world settings, there fore offering a non intrusive and cost-effective solution compared to traditional motion capture systems.

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Chapter 1

Introduction

Poor posture is one of the silent health issues in today's fast-moving digital world. It ranges from extended desk jobs to excessive screen time where a number of people unknowingly manage to acquire unhealthy postures that have worsened chronic back pains and joint discomforts, not to mention physical strain. Solutions addressing such issues must be innovative enough to fit into daily routines.

It's a new way of posture and gait analysis in real time using the advanced technology of OpenCV and Mediapipe for accurate classification. Using 1-D Convolutional Neural Network and IoT-enabled giving immediate feedback to them this proposed system not only identifies bad posture but also warns them to take corrective action right away.

With applications ranging from workplace ergonomics to fitness and rehabilitation, it so happens that this system thus propels the openness of possibilities for technology toward improving, at once, health and quality of life. This work sets forth the methodology, technical framework, and future directions enhancing the model to make it more easily accessible and impactful across settings.

Chapter 2

Literature Survey

2.1 Human Gait Modeling and Analysis Using a Semi-Markov Process With Ground Reaction Forces

This study advances real-time and easy-to-use gait analysis in a wearable system by employing force-sensitive smart insoles with FSRs to measure ground reaction forces when one is walking. It will discriminate and measure pathological gait patterns for clinic and home use, promising it would become an excellent diagnostic tool for abnormal gait, assessment of gait performance, and control of wearable assistive devices. It captures normal, age-related, and pathological gait patterns successfully and its quality is proved through comparison with a video-based gait abnormality rating scale. And, an interesting method of this technology bears promises for improvement in gait assessments in home and clinical environments.

2.2 Gait analysis using gravitational acceleration measured by wearable sensors

The study introduces a sensor system and algorithm designed to estimate lower limb gait posture, employing wearable sensor units that integrate a tri-axial acceleration sensor and three gyro sensors. These sensors, placed on the abdomen and lower limb segments, capture acceleration and angular velocity during walking. The method offers valuable quantitative data for gait analysis, making it applicable for both clinical diagnostics and continuous monitoring of daily human activities outside controlled environments. The algorithm's effectiveness was demonstrated through successful estimation of lower limb

posture and joint angles, providing a visual representation of varying walking patterns, thus enhancing gait assessment capabilities.

2.3 Evaluation of a Gait Analysis Tool Using Posture Estimation Technology in Clinical Rehabilitation

This study aimed to develop a gait analysis tool utilizing posture estimation technology to assess its ability to detect gait abnormalities in patients with knee osteoarthritis (OA). Using OpenPose, a technology that extracts key human body keypoints from video recordings, the researchers were able to analyze the gait patterns. The findings highlighted the tool's capacity to detect varus tilt (VT) in knee OA patients, with a notably larger change in knee varus angle during heel contact compared to healthy individuals. The study concluded that this gait analysis tool is an effective support tool for clinical evaluations, especially in identifying gait abnormalities in knee OA patients.

2.4 A model-based gait recognition method with body pose and human prior knowledge

It presents a novel model-based gait recognition approach, PoseGait, that uses human 3D pose estimation from images to better improve the performance in gait recognition. First, it uses OpenPose to estimate 2D poses from images and then extracts key 3D pose features like joint angles, limb lengths, and joint motions. Results show state-of-the-art performance of PoseGait and also robustness against variations in viewpoints and clothing. The method relies upon a rather low-dimensional feature set-only 14 body joints-but it nevertheless achieves high recognition rates and the ability to effectively contend with significant variations, especially in view angles.

2.5 Abnormal Gait Recognition Algorithm Based on LSTM-CNN Fusion Network

For the first time, human gait analysis is presented in the form of a combination of a wearable inertial measurement unit and a deep neural network for the detection and classification of abnormal gait patterns. This method relies on the use of a wearable IMU to collect gait data and then calculates the attitude angle for the gait data by applying the dynamic step conjugate gradient algorithm. It is easy to conclude that this dynamic

step size significantly enhances attitude measurement accuracy, while the LCSWnet framework classifies abnormal gaits quite well. To this aim, LCSWnet achieved 93.1 percent in its exciting accuracy above previous methods, leveraging temporal features and using a hybrid structure with an attention mechanism that leads to outstanding classification performance.

2.6 Measures of gait stability: performance on adults and toddlers at the beginning of independent walking

The purpose of this study is to evaluate the ability of different measures of gait stability to differentiate between unstable toddling and stable walking in adults. Ten toddlers, ten young adults, and 10 elderly individuals had their gait stability measured with two tri-axial inertial sensors for six months. The methods applied stride time variability, harmonic ratio, short-term recurrence quantification analysis, and multiscale entropy. The statistical comparisons between the different datasets were done using unpaired t-tests. This research is expected to prove the significance of gait stability understanding by applying these measures in toddlers who are considered antipriori unstable against subjects of young adults and the elderly. These findings emphasize the usefulness of certain gait stability indices for tracking developmental changes in toddlers and age-related changes in elderly individuals.

2.7 A vision based proposal for classification of normal and abnormal gait using RGB camera

This study aims to design a vision-based system that classifies normal and abnormal gait from the extracted RGB camera data. The method uses computer vision techniques applied to image preprocessing for gait analysis. The potential of the system is to achieve complete differentiation of normal and abnormal gait, which would categorize people who need early intervention and further medical assessment. The conclusion of the study is the creation of a general gait specification and an algorithm able to automatically extract those kinematic features that characterize the human gait, thus allowing effective classification of gait abnormalities.

2.8 Human Pose Estimation-Based Real-Time Gait Analysis Using Convolutional Neural Network

This system utilizes skeletal images derived from live video footage to detect gait abnormalities, with classifications based on key body points, such as the feet or toes. The method combines human pose estimation with a Convolutional Neural Network (CNN), processing skeletal images extracted from video frames for gait analysis. This approach offers a real-time, non-invasive solution to gait analysis using deep learning, providing high accuracy without the need for wearable sensors. The results are promising, with the system achieving an accuracy of 97.3 percent, demonstrating the effectiveness of combining pose estimation with CNN for accurate gait detection.

2.9 Human Gait Analysis Using OpenPose

It introduces a novel markerless approach to human gait analysis that is cost-friendly and friendly to use. The study used a mobile phone camera mounted on a tripod in capturing videos of subjects under various lighting conditions and attires. This simple setup is used by the method to provide an accessible mode of gait analysis devoid of such expensive pieces of equipment. Results According to findings, the proposed method based on OpenPose can correctly calculate knee extension angles with a percentage error of 16.78 percent under dim lighting and 13.43 percent in bright conditions. Therefore, it is a promising tool for practical gait analysis.

2.10 Human Gait Recognition Based On Motion Analysis

This work proposes a novel approach for automatic human gait recognition that combines computer vision techniques to analyze and classify human gait by integrating both static and dynamic characteristics into the feature extraction process. The method utilizes silhouette extraction for visual data, Discrete Cosine Transform (DCT) for feature analysis, and multi-class support vector machines (SVM) for classification. This approach achieves high recognition rates without requiring complex body modeling. The results are impressive, with the method achieving a 100 percent recognition rate on the Little and Boyd dataset, 90.2 percent on the CMU dataset, and 90.6 percent on the NLPR gait database, demonstrating its effectiveness and efficiency in distinguishing different human gaits using the extracted features.

Chapter 3

Methodology

3.1 Data Collection and Preparation

Importance of Data Collection

Success lies in the quality and diversity of training data as a prerequisite for machine learning models, especially for posture and gait analysis. Data collection for this project, therefore was carefully planned to ensure that the dataset would represent postures and variability in the real world. At the same time, the target was strong enough for changes in bodies, environments, and postural deviations yet kept simple enough for efficient processing.

3.1.1 Data Sources

Controlled image captures were employed with cameras placed at a fixed resolution for the purpose of data collection. The environment was controlled in a manner such that no aspects that might introduce confounding variables when taking the data were present. Examples of confounding variables include varied lighting conditions or distracting backgrounds.

1. Participants: A vast population with various characteristics, heights, weights, and build were considered. A diversified participant will ensure that the model generalizes very well across a wider audience.

2. Postures: There were four postures recorded, namely, good sitting, bad sitting, good standing, and bad standing. The classes were selected amongst the prevalent problems caused by postures in sedentary and workplace settings.

FIGURE 3.1: Posture Classes

3. Capture Environment: Images were taken in a brightly lit room where little background clutter is seen so that one can clearly identify body landmarks.

3.1.2 Collection Process

Data collection was done through the capturing of static images and live video frames of participants for each posture.

Static Frames : These were used as a way for pilot testing and training. About 30 images per class were taken, which gave an initial dataset of 120 labeled images.
Perspective Considerations: A fair amount of angles were employed to realistically mimic shooting photos with cameras in life situations: face on, wide-angle shot, and even subtle obliques.

3.2 Preprocessing of Data

Thus, preprocessing raw data is an important task that involves the preparation of raw data for training an effective machine learning model. It improves the quality of the dataset and its features to be consistent, standardized, and relevant to the model. Some of the major preprocessing steps within this methodology include:

3.2.1 Normalization of Landmark Coordinates

Thus, preprocessing raw data is an important task that involves the preparation of raw data for training an effective machine learning model. It improves the quality of the

dataset and its features to be consistent, standardized, and relevant to the model. Some of the major preprocessing steps within this methodology include

3.2.2 Handling Missing Data

Landmarks fail to detect on some images due to obstruction, incorrect pose, and so on, leading to missing or erroneous data points due to low quality of the image. Preprocessing of images involves correcting, as far as possible, incomplete or inaccurate key points or discarding them from the dataset altogether. All such frames in which body landmarks are poorly detected, and keys are missing or fall outside the expected bounds for certain frames, get filtered out. Such screening is really important as the wrong detection of key points can introduce noise using a single posture, thus possibly hindering discrimination between different postures.

3.3 Key-Point Detection and Feature Extraction

Basic to the process of understanding human posture based on body landmarks is a compound of two important preprocessing steps: key-point detection and feature extraction. These steps allow the system to capture spatial relationships among the parts of the body, used in the posture classification below.

3.3.1 Mediapipe Pose Estimation

Mediapipe, a framework developed by Google, is a state of the art tool for realtime computer vision tasks.

1. *Model Structure*
   Mediapipe's pose estimation model uses a combination of convolutional neural networks (CNNs) and heatmaps to detect body landmarks in 2D space. The model is trained to predict the location of each landmark (in terms of x, y, and z coordinates) from an input image.

2. *Landmark Detection*
   The model detects 33 body landmarks-shoulders, elbows, wrists, hips, knees, ankles, and spine-included. For each of these points, the system detects spatial coordinates; in case of a 2D image, this comes in the form of (x, y); for a 3D image, (x, y, z). These coordinates make for a skeletal representation of the subject

in an image-this is crucial for determining posture.

3. *Geometric Features*
   The most basic features used in describing posture are geometric relationships between the key body landmarks, including angles, distances, and alignments. These geometric features capture relative positioning and orientation of different parts of the body, which are the key indicators of whether a posture is good or bad.

   Joint Angles: This is one of the main features. For example,

   This angle between spine and hips indicates the torso's tilt. The angle between the knees and thighs helps evaluate whether the posture is correct or not. The shoulder-to-hip angle is especially relevant for finding out if the upper body is in alignment or slouching. These angles are found by considering the coordinates of three or more connected landmarks and using simple trigonometric formulas in order to obtain the angle which forms at a joint.
   Distances: Useful features also include distances between major landmarks.
   Distance between shoulders and hips may be useful in establishing whether the upper body is well structured or slumped. Distance between the hands and body may help in determining if the arms are relaxed or tense, indicating possible poor posture.
   Body Segment Ratios: The algorithm calculates ratios of various distances between parts of the body. For example, a ratio of distance between shoulders and hips to the overall height may assist in building information on the posture and alignment of the subject.

4. *Feature Vector Representation*
   Once extracted, the features are arranged into a numerical format compatible with the learning model used in the machine. These features are generally stored as one-dimensional array for each frame of the video. For example, the value contained in the array represents a specific feature like the angle of a joint or distance.

3.4 Model Architecture

The model architecture for posture classification leverages a 1D Convolutional Neural Network (CNN), chosen for its ability to capture local dependencies in sequential or structured data, such as the spatial relationships between body landmarks.

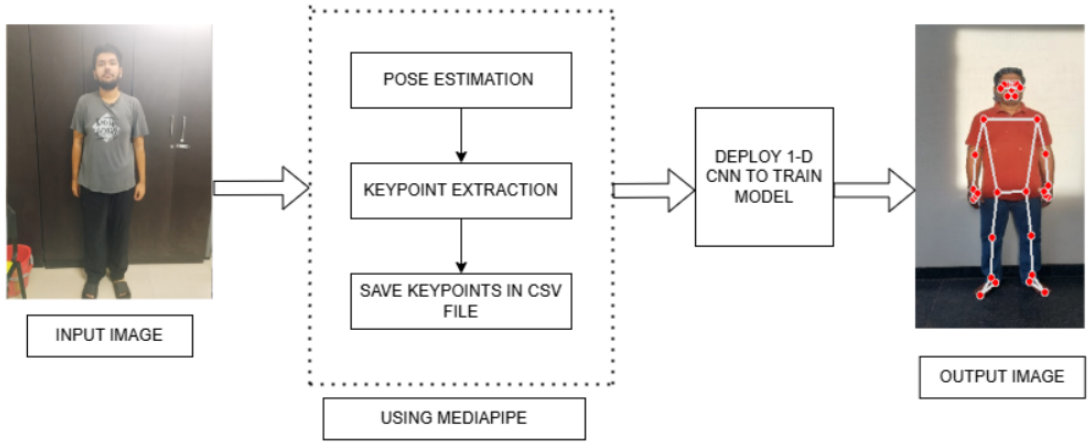


FIGURE 3.2: Input Flowchart

3.4.1 Model Overview

We construct the model to classify human posture into four categories-good sitting, bad sitting, good standing, and bad standing. The challenge mainly is to take data coming from body landmarks representing the human pose and translate it into meaningful posture predictions. The architecture itself will be based upon a 1D CNN that really is well suited to tasks involving either sequential or structured data, just like our key point features.
Input: The model input is a tensor containing the 3D coordinates of 33 body landmarks (x, y, z coordinates), so the tensor must be of shape (33, 3).
Output : The model gives out a 4class classification, of which each of the class represents any of the posture categories, such as good sitting, bad sitting, good standing and bad standing. Usually, this output is generated by softmax layer which gives out output as the probabilities of each class.

3.4.2 Model Structure and Layers

The architecture of the 1D CNN can be broken down into several layers, each with a specific role in feature extraction and classification.

3.4.3 Input Layer

Shape: The input to the model is a tensor with a shape of (33, 3), which corresponds to the 33 key points detected by Mediapipe. Each key point has 3 coordinates: x, y, and z.
Input Tensor: (33, 3) $\rightarrow$ 33 body landmarks with 3 spatial coordinates each (x, y, z).
The input layer passes this raw data directly to the convolutional layers.

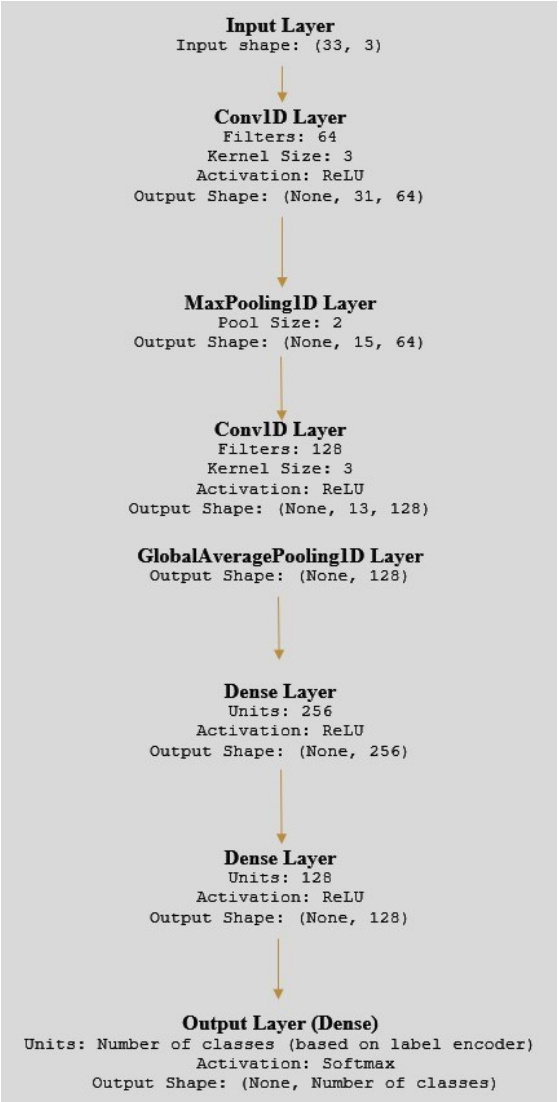


FIGURE 3.3: 1-D CNN Workflow

3.4.4 Convolutional Layers

Convolutional layers are designed to detect spatial hierarchies in the input data by learning local patterns or features. In posture classification, these layers learn how adjacent body parts work together or how to align with the overall body. Convolutional layers apply a set of filters (kernels) over the input data to extract important features like angles, distances, and alignment ratios.

Layer 1: First Convolutional Layer Number of Filters: 64 Kernel Size: 3 (small kernels to catch local features of body landmarks) Activation Function: ReLU (Rectified Linear Unit)

Output: The first convolutional layer applies 64 filters on the input tensor. This will pull out low-level features such as basic movement patterns and orientations of body segments.

Layer 2: Second Convolutional Layer Number of Filters: 128 Kernel Size: 3 (once more, a small kernel size to focus on the local patterns) Activation Function: ReLU Output: The second convolutional layer will use 128 filters to extract more complex features. It should take low-level patterns captured by the first convolutional layer and combine them into higher-level abstractions. Stride and Padding: For the second convolutional layer, the stride is set once more to 1 with padding = 'same' to preserve the size of the output feature map.

Max-Pooling Layers

The max-pooling layers reduce the dimension of the data with emphasis on the larger features. They reduce the feature maps by choosing the largest value out of a set of neighboring values, like a 2x2 area to facilitate the computation reduction and avoid overfitting.

Max-Pooling Layer 1: Pool Size: 2x1 applied to the feature maps produced by the convolutional layers.

Stride: 2 (this effectively reduces the spatial dimensions of the data, which also scales the number of parameters to be learned in the following layers)

Pooling provides some form of translational invariance, thus allowing the model to recognize postures even when the body position was slightly shifted.

Fully Connected Layers

Fully connected layers are used to aggregate the features and generate the final prediction after extracting the features. They engage in high-level reasoning with low-level and mid-level features derived through previous layers.

Fully Connected Layer 1: Units: 256 (neurons)

Activation Function: ReLU The dense layer brings out all the feature extractions and helps the model make sense of the high-level patterns, thus preparing the data for the classification. It gives non-linearity with the ReLU activation to help the network learn more complex relations.

Fully Connected Layer 2 Units: 128 units. We will have fewer neurons in this layer compared to the previous one since we want to keep in mind that we are gradually reducing the dimensionality.

Activation Function: ReLU Purpose: This layer further processes the feature representation in order to refine the output, thus making the model more discriminative in the different posture categories.

Output Layer Units: 4 (as this is the one corresponding to the 4 posture categories)

Activation Function: Softmax

In the output layer, softmax activation is applied to compute a probability distribution over the four possible classes. The output values will add up to 1 through softmax: that would allow the model to predict a probability for each class; it will then output the class that has the highest probability.

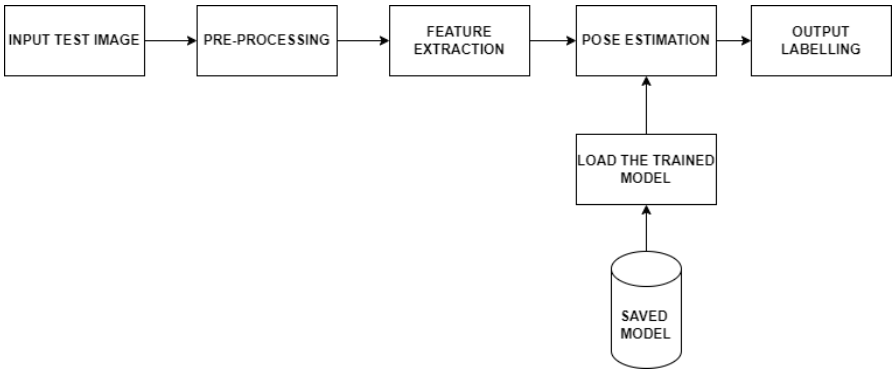


FIGURE 3.4: Workflow

3.5 Real-Time Implementation

In real-time implementation, the posture classification model trained will be applied to the video data in real time, where the system will be able to detect and classify the posture of a person as he moves and changes position. It can then lead to dynamic feedback, such as posture monitoring and correction, for applications in many fields, including assessment for ergonomics, health monitoring, and physical therapy.

3.5.1 System Overview

The system integrates the posture classification model into a video stream, for example from a webcam or camera, and applies the model frame by frame to analyze the user's posture. The components of the system are as follows:

1. Camera Feed: This includes the capturing of live video frames.The use of a pose estimation model such as Mediapipe to estimate key landmarks of the person in the frame.

2. Posture Classification: Classifying the posture by passing the detected key points through the trained 1D Convolutional Neural Network (CNN) model. Real-time posture feedback will be conveyed to the user through overlaying posture labels onto the live feed or through some visual/haptic alerts.

3.5.2 Camera Feed Capture

The first step in live implementation is the capture of video from a camera. This can be facilitated using OpenCV, which contains tools that do handle video input from webcams, process video frames, and output results.

- Video Stream: The feed from a camera is accessed using OpenCV's `cv2.VideoCapture()` function, which captures video frames in real time. These then form the continuous source of using them for processing.
- Frame Rate: The system updates frames at a reasonable frame rate in order to give near-instant feedback.

3.6 EsP-32 Implementation

In the case of real-time posture monitoring, a feedback system combines with an ESP32 with a software-based posture classification system. The use of physical, real-time feedback adds another dimension to the interaction with the user based on sensory output. The key idea here is to make posture correction actionable because there would be an alert when users assume a bad posture, thus allowing corrective actions to be taken promptly.

3.6.1 System Overview and Components

The real-time physical feedback for the users when poor posture is identified will be offered by the IoT-enabled system. .
An IoT device normally in the form of a microcontroller such as the ESP32 controlling an actuator in this case, a vibrating motor which provides the haptic feedbacks to the human user when bad posture has been detected. The ESP32 is a popular microcontroller used in IoT projects due to its Wi-Fi and Bluetooth capabilities. It is affordable, power-efficient, and flexible, making it ideal for real-time applications where wireless communication is needed.
The ESP32 microcontroller is used to receive the posture classification output from the machine learning model running on a PC or edge device, and then trigger feedback mechanisms like a vibrating motor.

3.6.2 System Workflow

he The following is a step-by-step decomposition of the workflow for the IoT-enabled feedback system:
Capture Video : Webcam or camera captures the user's image.
Pose Estimation: Mediapipe's pose estimation model detects the key landmarks in the captured frame and outputs their coordinates.
Posture Classification: The system processes the body landmarks using the trained CNN

FIGURE 3.5: Realtime Implementation

model to classify the user's posture.

Feedback Decision: Whenever the posture is rated as bad like "bad sitting," the system must generate some signal towards the ESP32.

Sending Signal: It receives the signal through Wi-Fi or Bluetooth and the haptic feedback mechanism of the vibrating motor is activated in the IoT device.

Giving it to the user: A vibration is produced and a message to adjust the posture is sent.

Real-Time Monitoring: It keeps monitoring the posture of the user in real-time and provides feedback when needed.

Chapter 4

Results and Discussions

Objective

This work has the potential to build a hybrid approach of computer vision, machine learning, and IoT for real-time posture monitoring and correction. The system was able to classify postures with high precision in real time and further provided actionable feedback to the users through an IoT-enabled device. The solution can be transferred to the real world-the home, office, and healthcare environment-to prevent musculoskeletal disorders and assure better health.

The system will thus provide an interactive solution for posture correction with a user-friendly approach, integrating into a system feedback provided by IoT, such as wearing devices with haptic feedback. The combination of machine learning with IoT may revolutionize the way people are monitored in keeping the habit of healthy postural habits. While good on the test set, testing the model in real life conditions is far more crucial to

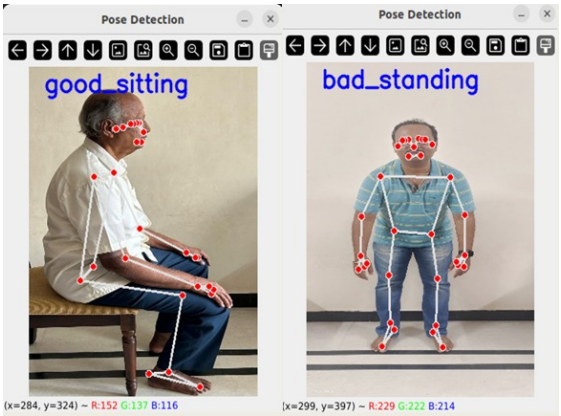
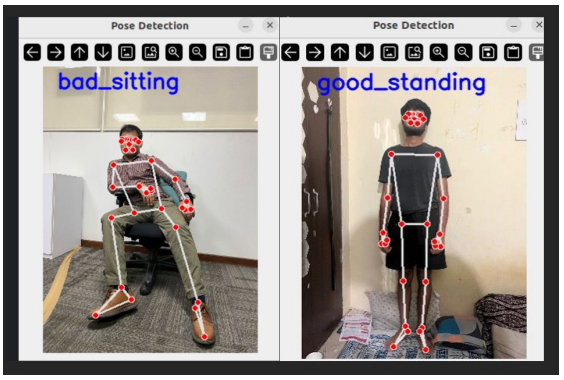


Figure 4.1: RESULTS



proving true robustness. Here, the system was tested in many different lighting conditions, such as low light and high light, simulating real-world conditions that might affect the model in action, such as poor visibility or shadows. It performed well in better-lit conditions but lost a bit of its accuracy at lower light, or where the landmark view was partially obstructed-for example, by a hand covering part of the torso. The model was tested with participants representing various demographics in terms of age, gender, and body type to ensure that it generalizes well across different users. While very good for the majority of people, this model did demonstrate some level of performance degradation on people with atypical body proportions like those who are extremely tall or short and for users with non-standard sitting postures like cross-legged settings. These will require further dataset expansion and model refinement.

Model Evaluation confirmed that the posture classification system performs suitably in real-world applications with high levels of accuracy, precision, and recall. This model is robust but provides room for improvement in handling different users and environments. In the future, it may be possible to improve on this by expanding the dataset, by detecting landmarks even under adverse conditions, and fine-tuning the model to even more sensitively capture posture deviations.

Chapter 5

Conclusion and Future Scope

This study proposed a real-time gait classification model based on 1D-CNN architecture. Through efficient key-point extraction and classification using CNN, model was able to classify four different postures. Results indicate the model's precision and its usefulness for postural correction and control in real time. The approach proposed, although useful, has room for improvement on the existing method. In the future researches, aiming to minimize the computational cost, enhance the resilience against light conditions, and look into extending to bigger sizes data sets In addition to this, it will be more interesting to see how this model will play out when fused with a wearable or mobile application for better usability. This multimodal approach would allow for a more nuanced understanding of body mechanics, capturing subtle movements that might be missed through visual data alone. The model's ability to monitor and correct posture in real time could have profound implications for workplace health, especially in environments where employees are sedentary for long periods.

In the upcoming work, we will focus on the improvement of the dataset in terms of quality and quantity by capturing images in more diverse conditions and involving different types of bodies. Furthermore, the use of transfer learning through the application of previously trained models is expected to enhance the quality of generalization. In addition, looking into the methods of 3D pose estimation would help tackle the problem of occlusion and improve the correctness of the posture classification even further. We also working on creating web based platform for bad posture recognition mainly focusing for corporates. While the model currently performs well with the dataset used, expanding to larger datasets will improve its generalizability and ability to handle diverse postures. Future research could involve gathering extensive and varied datasets from different age groups, body types, and activity levels, enhancing the model's versatility and resilience.

Chapter 6

Appendix

Preprocessing: For normalizing and preprocessing landmark coordinates

Model Architecture: Implementation of the 1D CNN model using TensorFlow

Real-time Implementation: Camera setup and frame processing functions

ESP32 Integration: WiFi setup and feedback control

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